Update Notice

As we described before, duplicate trial entries resulting from repeated runs within the same recording session were identified and removed. These duplicates were strictly confined to within-session repetitions and did not occur across different sessions or participants. Following this cleanup, the updated dataset comprises 23,382 trials (corresponding to 93,528 one-second windows).

Consequently, all relevant experiments were re-evaluated. Table 6 which reported the results for the five-fold cross-trial motor imagery (MI) classification is updated here, while other tasks—specifically cross-session classification and FMA score regression—remain unaffected. Models marked with † indicate updated performance metrics compared to the original reported values. Currently, EEG-Conformer achieves the highest cross-trial accuracy at 70.48%, surpassing ATCNet, which previously yielded the top performance.

Table 6 (Section "Technical Validation") Performance comparison of eight EEG decoding models under the 5-fold cross-trial data-splitting strategy. † indicates results changed from the previous report; bold indicates the best performance.

Classifier	Accuracy (mean $\pm$ SD, %) $\uparrow$	F1 $\uparrow$	Kappa $\uparrow$
TSLDA + DGFMDRM	53.01 ± 2.13	0.512	0.060
FBCSP + SVM	56.32 ± 4.48	0.561	0.126
TWFB + DGFMDRM	56.91 ± 3.37	0.567	0.138
EEGNet	61.93 ± 3.18	0.600	0.238
EEGConformer†	$\mathbf{70.48 \pm 7.23}$	**0.599**	**0.263**
IFNet†	67.63 ± 7.50	0.592	0.232
ArjunViT†	64.90 ± 6.02	0.511	0.122
ATCNet†	67.66 ± 7.72	0.579	0.220